

Full asymptotic expansions for partition sums of acyclic orientations

1 October 2026

Abstract

We obtain the full prefactors and arbitrary finite orders of the asymptotic expansions for OEIS A372395 and A370613, which sum acyclic orientations of complete multipartite graphs over unrestricted and distinct-part integer partitions. The normalized counts have the respective forms $A_+ n^{-1} e^{C_+ \sqrt{n}}$ and $A_- n^{-3/4} e^{C_- \sqrt{n}}$, with corrections in powers of $n^{-1/2}$. The constants $C_{\pm}$ were previously identified by Sun. The amplitudes include a nontrivial contribution from a tilted Gamma integral. A uniform quadratic majorant supplies relative-precision tail control despite the signed polynomial integral representation. We give a finite coefficient algorithm, explicit first and second corrections, reproducible exact checks through $n = 300$, and controlled Lambert- W inverse expansions. The result is an all-orders Poincaré expansion; it does not assert an exponentially complete transseries.

1 Results and relation to prior work

For an integer partition λ of n , let K_λ be the complete multipartite graph with part sizes $\lambda_1, \lambda_2, \dots$. Define

$$B_+(n) = \sum_{\lambda \vdash n} \text{AO}(K_\lambda), \quad B_-(n) = \sum_{\substack{\lambda \vdash n \\ \lambda \text{ distinct}}} \text{AO}(K_\lambda).$$

These are OEIS A372395 and A370613 [1, 2]. Each partition profile occurs once; there is no additional multinomial choice of vertex sets. We set both empty counts equal to one.

Sun [4] derives the exact Gamma representation used below and proves $\log(B_{\pm}(n)/n!) \sim C_{\pm} \sqrt{n}$. His Section 11 explicitly asks for the prefactors and complete expansions. The new refinement here is at relative precision: the amplitudes, every algebraic correction, and the associated inverse. The main additional estimate is a uniform quadratic majorant which removes the large-part cutoff without losing an $e^{o(\sqrt{n})}$ factor.

Use $\eta = +1$ for unrestricted partitions and $\eta = -1$ for distinct parts. For real c in its natural domain, put

$$\rho_\eta(x; c) = \frac{1}{e^{cx+x^2/2} - \eta}, \tag{1}$$

$$J_\eta(c) = -\eta \int_0^\infty \log(1 - \eta e^{-cx-x^2/2}) dx, \tag{2}$$

$$M_j(c) = \int_0^\infty x^j \rho_\eta(x; c) dx, \quad j \geq 1. \tag{3}$$

The Bose case requires $c > 0$; the Fermi case allows every real c . There is a unique $c = c_\eta$ such that $M_1(c) = 1$. Indeed M_1 is continuous and strictly decreasing, with limits infinity and zero at the

two endpoints of its domain. In the Bose case the divergence as $c \downarrow 0$ is logarithmic at the origin; in the Fermi case it follows from the interval where $cx + x^2/2 < 0$ as $c \rightarrow -\infty$.

All unlabelled M_j, J below are evaluated at this saddle unless their argument is shown. Set

$$C = c + J(c), \quad V = J''(c) = \int_0^\infty x^2 \rho(1 + \eta \rho) dx > 0, \quad (4)$$

$$E_0 = \exp\left(\frac{1}{2} - \frac{M_3}{3} + \frac{M_2^2}{8}\right), \quad (5)$$

$$p_+ = 1, \quad A_+ = \frac{\sqrt{c_+} E_{0,+}}{2\pi\sqrt{V_+}}, \quad p_- = \frac{3}{4}, \quad A_- = \frac{E_{0,-}}{2\sqrt{\pi V_-}}. \quad (6)$$

Theorem 1.1. *For each $\eta \in \{+1, -1\}$ there are real constants $a_{\eta,j}$, with $a_{\eta,0} = 1$, such that for every fixed integer $L \geq 0$,*

$$B_\eta(n) = A_\eta n! n^{-p_\eta} e^{C_\eta \sqrt{n}} \left(\sum_{j=0}^L a_{\eta,j} n^{-j/2} + O_L(n^{-(L+1)/2}) \right). \quad (7)$$

Section 7 gives a finite algorithm for every coefficient.

	Unrestricted	Distinct parts
c_η	0.764996442279544302	-0.323697314095031872
C_η	2.158752005657785532	0.905729821720199018
A_η	0.164817523964420504	0.217999212918202522
$a_{\eta,1}$	-0.307565333551633697	0.230950213208614981
$a_{\eta,2}$	0.018726632258564924	-0.004247066120392843

The power series in (7) is asymptotic in the finite-order sense. No convergence of its infinite sum is needed or claimed. Numerical constants in this table are evaluations of the integral and finite coefficient formulas, rather than fitted parameters.

[write] **Added 2 October 2026, batch 77.** This report is a single manuscript, printed in full: manuscript 59 of batch 77 of ProveIt's incoming-reports intake, delivered as the archive `orientation-partition-asymptotics.zip` in commit 096ee7b87 (main file `orientation-partition-asymptotics.tex`, now `article.tex`) and placed in commit d0e6008d9 in the research-report collection, among the OEIS-sequence asymptotics reports. The manuscript is dated 1 October 2026, its author line is empty, and the delivery names no author and no tool. It names no ProveIt commit and continues no repository path, so it has no pin. Nothing in this report is formalized in Lean or Rocq, and placement in the collection confers no formal status on any statement here. The writing step prefixed every label with `aop:`, and added these [write] notes and two references [5, 6]; no statement, proof or number of the manuscript was altered.

Credit and scope. The exact Gamma representation (9) and the constants $C_\pm$ (with $\log(B_\pm(n)/n!) \sim C_\pm \sqrt{n}$) are Sun's [4]; the question answered here, the prefactors and the complete expansion, is the one his Section 11 poses. The expansion is asymptotic to every fixed order in the Poincaré sense; no exponentially complete transseries, exponentially small sector or summation prescription is asserted (Section 12). The numerical tables are evaluations and residual checks, not interval certificates.

Neighbouring reports. No report of the collection treats acyclic orientations, chromatic polynomials of complete multipartite graphs, A372395 or A370613; the Touchard and Stirling polynomials used here appear elsewhere in the collection only as classical tools. The batch-77 reports `a301981-unitary-divisor-partitions` and `a174065-radix-layer-partitions`, and the earlier partition-product reports such as `a022629-distinct-partition-norms`, also apply saddle-point analysis to weighted partition products, but to other products; no result is shared.

Table 1: [write] Notation of this report (added 2 October 2026, batch 77).

symbol	meaning here (and what it is <i>not</i>)
η	the sign +1 (unrestricted, Bose) or -1 (distinct, Fermi); not a small parameter. The small parameter is $\varepsilon = 1/s = n^{-1/2}$.
A_η, A, a	A_η the amplitudes (6); $A = (Q-n)/2$ and $a = A/n^2$ in the proof of Theorem 3.2; $as \leq k \leq bs$ in Section 6; a, b, d the inverse terms (45); $a_{\eta,j}$ the expansion coefficients.
$B_\eta(n), B, \mathbf{B}_j$	the two sequences (A372395, A370613), with cutoff and chromatic variants $B_{\eta,K}, B_{\eta,v}$; B a root-sum quantity in the proof of Theorem 3.2; $\mathbf{B}_j, \mathbf{B}_r(v)$ Bernoulli numbers and polynomials.
C_η, c_η, C_0	$C_\eta = c + J(c)$ the growth constant and c_η the saddle; C_0 the majorant constant of Theorem 3.2 and, separately, an angle constant C_0/s in Section 6 (with c_0/s); C_1, C_2 bound constants.
$d_\eta(c), d_0$	$d_\pm(c)$ the constants of (28); $d_0(x; c) = f_0 - f_{\text{std}}$ a <i>function</i> in Section 5; $d_0 = \log(A\sqrt{2\pi})$ a <i>constant</i> in (43); d, d', d_2, d_3 positive constants; d also the inverse term (45).
$E_0, E_j, \mathbb{E}$	E_0 the amplitude factor (5); $E_j(c)$ Euler–Maclaurin endpoint terms in (32); $\mathbb{E}$ an expectation.
$H_j, H_v, H_{\pm,n}$	$H_j(c, z)$ the product coefficients (28); $H_v(G) = (-1)^{ V(G) } \chi_G(-v)$ the chromatic evaluation (Section 11); $H_{\pm,n}(t)$ the Touchard coefficient polynomials (Section 10).
K, K_j, K_λ	$K = \lfloor s \log n \rfloor$ the part cutoff; K_j the cumulant integrals (Section 8); K_λ the complete multipartite graph.
L	a fixed order in Theorem 1.1; the largest part $\max \lambda$ in Section 4 and Corollary 11.1; $L = \log Y$ in Section 9.
ℓ_j, ℓ	$\ell_j(x, z)$ the root-expansion polynomials (24)–(25); $\ell = \log N$ in Section 9.
M_j, M	the moment integrals $M_j(c)$; $M = \max_i \lambda_i$ in the proof of Theorem 3.2 and Corollary 11.1.
$P_k, p_r, p(n), p_\eta$	$P_k(t)$ the signed Stirling polynomials, $R_k = P_k/t^k$; $p_r(k)$ their root power sums; $p(n)$ the partition count; $p_\eta \in \{1, 3/4\}$ the power of n ; P a polynomial bound in Section 7.
$Q, Q_k, \mathcal{Q}$	$Q(\lambda) = \sum \lambda_i^2$; $Q_k(t) = \sum_j S(k, j) t^j$ the Touchard polynomials (Section 10); $\mathcal{Q}$ the formal exponent (36).
$R, R_\eta(n), r$	R a truncation order in (23), (27); $R_\eta(n)$ the exact ratio and $T_\eta(n)$ the scaled residual (Section 10); r_η the logarithmic coefficient, r_j the inverse polynomials, $r_{\max}$ the largest root.
T, τ	$T \sim \text{Gamma}(n+1, 1)$ the Gamma variable; $\tau(Y)$ the integer threshold.
$F, \mathcal{F}_\eta, G$	$F_{\eta,K}(q, t)$ the cutoff product; $\mathcal{F}_\eta$ the positive weighted model (Section 4); $F(x)$ the smooth interpolant and $G(x)$ a smooth realization (Section 9).
$\alpha, \alpha_0; \beta, \beta_j$	$\alpha = M_2/2$, $\alpha_0 = 1/2 - p$; $\beta \in (3/4, 4/5)$ the cutoff exponent, β_j the logarithmic coefficients (43).
N	$N = L/W(L/e)$ the Lambert core of the inverse; not the index n .

Notation. Several letters carry more than one meaning. Each is printed as delivered; Table 1 fixes the meanings, with the tempting false readings. No symbol was renamed.

2 Exact representations and their signs

Let $S(k, j)$ denote the Stirling numbers of the second kind, and define

$$P_k(t) = \sum_{j=0}^k (-1)^{k-j} S(k, j) t^j, \quad R_k(t) = \frac{P_k(t)}{t^k}.$$

Their exponential generating function is

$$\sum_{k \geq 0} P_k(t) \frac{z^k}{k!} = \exp(t(1 - e^{-z})). \quad (8)$$

A proper coloring partitions each independent vertex part into nonempty monochromatic blocks, and different parts must use different colors. Thus, writing q^j for a falling factorial,

$$\chi_{K_\lambda}(q) = \sum_{j_1, \dots, j_r} q^{j_1 + \dots + j_r} \prod_i S(\lambda_i, j_i).$$

Stanley's identity $\text{AO}(G) = (-1)^{|V(G)|} \chi_G(-1)$ [3], followed by the Gamma integral for $(j_1 + \dots + j_r)!$, proves

$$\text{AO}(K_\lambda) = \int_0^\infty e^{-t} \prod_i P_{\lambda_i}(t) dt. \quad (9)$$

This is also Sun's exact multipartite integral. Its integrand can have either sign. Infinite products formed from it are interpreted coefficientwise, not as globally convergent analytic products.

For a finite part cutoff K , put

$$F_{\eta, K}(q, t) = \prod_{k=1}^K (1 - \eta q^k R_k(t))^{-\eta}.$$

If $B_{\eta, K}$ denotes the sum restricted to largest part at most K , then, for $T \sim \text{Gamma}(n+1, 1)$,

$$\frac{B_{\eta, K}(n)}{n!} = \mathbb{E}_T[q^n] F_{\eta, K}(q, T). \quad (10)$$

Coefficient extraction comes first. Every selected profile has total size n , so the removed powers of t contribute exactly t^n , producing the Gamma density in (10). This finite identity makes no positivity assumption about the polynomials on the whole half-line.

3 A uniform quadratic majorant

Lemma 3.1. *The roots of P_k are distinct and nonnegative, including zero. If $p_r(k)$ is their r th power sum, then it is a polynomial in k of degree at most $r+1$, determined by*

$$p_r(k+1) - p_r(k) = -r[u^r] \log \left(1 - ku - \sum_{j \geq 1} p_j(k) u^{j+1} \right), \quad p_r(0) = 0. \quad (11)$$

In particular, the largest root is $O(k^{5/4})$.

Proof. The Stirling recurrence gives $P_{k+1}(t) = t(P_k(t) - P'_k(t))$, with $P_1(t) = t$. At consecutive roots of P_k , the values of $P_k - P'_k$ alternate in sign. There is one root in each intervening interval and one to the right of the largest root; all are positive. Multiplication by t completes the induction.

Writing $\mathcal{R}_k(u) = \prod_i (1 - r_i u)$, we have $\mathcal{R}_{k+1} = (1 - ku)\mathcal{R}_k + u^2 \mathcal{R}'_k$. Taking logarithms yields (11). Only p_j with $j < r$ is needed to compute the right side. Its degree in k is at most r , so finite summation proves the degree bound. Explicitly,

$$p_1(k) = \frac{k(k-1)}{2}, \quad p_2(k) = \frac{k(k-1)(4k-5)}{6}, \quad (12)$$

$$p_3(k) = \frac{k(k-1)(9k^2 - 25k + 18)}{8}, \quad (13)$$

$$p_4(k) = \frac{k(k-1)(64k^3 - 291k^2 + 459k - 251)}{30}. \quad (14)$$

All roots are nonnegative, so their maximum is at most $p_4(k)^{1/4}$. $\square$

Theorem 3.2. Fix $3/4 < \beta < 4/5$. There is a constant C_0 such that, uniformly over partitions λ of n whose largest part is at most n^β ,

$$\frac{\text{AO}(K_\lambda)}{n!} \leq C_0 \exp\left(-\frac{Q(\lambda) - n}{2n}\right), \quad Q(\lambda) = \sum_i \lambda_i^2. \quad (15)$$

The central integral also has Gaussian suppression in $z = (t - n)/\sqrt{n}$, uniformly over these profiles.

Proof. Put

$$A = \frac{Q - n}{2}, \quad B = \sum_i \frac{\lambda_i(\lambda_i - 1)(4\lambda_i - 5)}{12}.$$

Lemma 3.1 and $\beta < 4/5$ imply that every relevant root is $o(n)$. All factors are therefore positive on $[n/2, 2n]$. The exact inequality $\log(1 - v) \leq -v - v^2/2$ gives

$$\prod_i P_{\lambda_i}(t) \leq t^n \exp(-A/t - B/t^2). \quad (16)$$

Weighted Cauchy–Schwarz, followed by a termwise inequality, gives

$$\frac{A^2}{n} \leq \sum_i \frac{\lambda_i(\lambda_i - 1)^2}{4} \leq B.$$

The difference in the second inequality for a part k is $k(k - 1)(k - 2)/12 \geq 0$.

Let $x = t/n$, $a = A/n^2$ and $g(x) = \log x - x + 1$. Relative to $n^n e^{-n - A/n}$, the integrand is bounded by

$$\exp\{n[g(x) + a(1 - 1/x) - a^2/x^2]\}.$$

For $x \leq 1$ the two a terms are nonpositive. For $x \geq 1$ their supremum over $a \geq 0$ equals $(x - 1)^2/4$. On $[1/2, 2]$ there is a constant $\delta > 0$ with

$$g(x) + (x - 1)^2/4 \leq -\delta(x - 1)^2. \quad (17)$$

Indeed its derivative is $(x - 1)(1/2 - 1/x)$; its value at 2 is $\log 2 - 3/4 < 0$, and division by $(x - 1)^2$ has limiting value $-1/4$ at 1. Integrating the resulting Gaussian bound and using Stirling’s formula proves (15) for the central integral. The same calculation gives the pointwise factor $e^{-\delta z^2}$.

It remains to control signs outside that interval. Unsigned first-kind Stirling numbers dominate the second-kind ones: specifying a cyclic order on each block of a set partition produces a permutation. If $M = \max_i \lambda_i$, then, for $t \geq 0$,

$$|P_k(t)| \leq \sum_j S(k, j) t^j \leq t(t + 1) \cdots (t + k - 1) \leq (t + M)^k.$$

Hence the absolute tail integral is bounded by

$$\int_{t \notin [n/2, 2n]} e^{-t} (t + M)^n dt = n! \exp(-\Omega(n) + O(M)).$$

Substitution of $t + M$ and elementary Gamma Chernoff bounds justify this uniformly for $M = o(n)$. Since $A/n \leq M/2$, the tail is negligible relative to $n! e^{-A/n}$. Positivity of the orientation count and the triangle inequality complete the proof. $\square$

4 Relative precision cutoff removal

Write $s = \sqrt{n}$ and choose $K = \lfloor s \log n \rfloor$. For the intermediate range $K < \max \lambda \leq n^\beta$, the majorant reduces the sum to a positive weighted partition model, with

$$w_k = e^{-k^2/(2n)}, \quad q_0 = e^{-c/s}, \quad \mathcal{F}_\eta(q_0) = \prod_{k \geq 1} (1 - \eta w_k q_0^k)^{-\eta}.$$

For the unrestricted model all geometric ratios are below one. In the distinct model some ratios exceed one, but each factor remains positive and defines a Bernoulli distribution. A union bound bounds the weighted coefficient containing a part greater than K by

$$q_0^{-n} \mathcal{F}_\eta(q_0) \begin{cases} \sum_{k > K} \frac{w_k q_0^k}{1 - w_k q_0^k}, & \eta = +1, \\ \sum_{k > K} w_k q_0^k, & \eta = -1. \end{cases} \quad (18)$$

The sums are bounded by $\exp(-\Omega(\log^2 n))$ times a polynomial. The Euler–Maclaurin calculation below, already at leading order, gives $q_0^{-n} \mathcal{F}_\eta(q_0) = \exp(Cs + O(\log n))$. Thus this range is negligible relative to e^{Cs} times any fixed inverse power of n .

For the remaining range let $L = \max \lambda > n^\beta$. Splitting every other part into singleton parts adds edges, so deletion–contraction implies

$$\text{AO}(K_\lambda) \leq \text{AO}(K_{L,1,\dots,1}) = (n-L)!(n-L+1)^L.$$

The formula follows by ordering the singleton vertices and placing each of the L labeled remaining vertices into one of the $n-L+1$ gaps. There is an exact bound valid for every $0 \leq L \leq n$:

$$\frac{(n-L)!(n-L+1)^L}{n!} = \prod_{j=1}^L \left(1 + \frac{j-1}{n-L+1}\right)^{-1} \quad (19)$$

$$\leq \exp\left(-\frac{L(L-1)}{2n}\right). \quad (20)$$

Use $\log(1+x) \geq x/(1+x)$ and $n-L+j \leq n$. Since $2\beta-1 > 1/2$, multiplication by the standard bound $p(n) \leq e^{O(\sqrt{n})}$ leaves a negligible tail.

The Gamma variable can also be restricted to

$$|T - n| \leq s \log^2 n. \quad (21)$$

For profiles with largest part at most K , the pointwise form of (17) gives suppression $e^{-\delta z^2}$ on $[n/2, 2n]$, in addition to the weight $e^{-(Q-n)/(2n)}$. Summing these fixed-total weights gives the same positive weighted-partition coefficient as above. The complement of (21) is therefore smaller than every algebraic order on the main scale. Outside $[n/2, 2n]$ the absolute signed-tail estimate applies. This argument uses positivity only where it was proved.

5 The local product and Euler–Maclaurin expansion

Set $\varepsilon = 1/s$ and $T = s^2 + sz$. On the cutoffs already justified,

$$\log R_k(T) = - \sum_{r \geq 1} \frac{p_r(k)}{r T^r}. \quad (22)$$

This convergent root expansion has a useful finite remainder bound: after $r = R$, its absolute remainder is at most

$$\frac{p_{R+1}(k)}{(R+1)T^{R+1}(1-r_{\max}/T)}. \quad (23)$$

Here $r_{\max}/T = o(1)$ uniformly and $p_{R+1}(k) = O_R((1+k)^{R+2})$. Consequently, with $x = k\varepsilon$,

$$\log R_k(s^2 + sz) = -x^2/2 + \sum_{j \geq 1} \varepsilon^j \ell_j(x, z)$$

is a controlled arbitrary finite expansion. Each ℓ_j is a polynomial vanishing at $x = 0$. The first two are

$$\ell_1 = x/2 + zx^2/2 - x^3/3, \quad (24)$$

$$\ell_2 = -zx/2 - z^2x^2/2 + 2zx^3/3 + 3x^4/4 - 3x^4/8. \quad (25)$$

Define f_j by formal coefficient extraction in

$$-\eta \log \left(1 - \eta \exp \left[-cx - x^2/2 + \sum_{j \geq 1} \varepsilon^j \ell_j(x, z) \right] \right) = f_0(x; c) + \sum_{j \geq 1} \varepsilon^j f_j(x; c, z). \quad (26)$$

For $j \geq 1$, f_j is smooth at zero, including in the Bose case. The m th derivative of the outer logarithm has a pole of order at most m , whereas a product of m perturbation factors vanishes to at least order m . At infinity all relevant derivatives decay as a Gaussian times a polynomial. These properties persist for c in a sufficiently small complex neighborhood of its real saddle, with polynomial bounds in z .

In the Fermi case f_0 is smooth and $f_0(0) = \log 2$. In the Bose case subtract $f_{\text{std}}(x; c) = -\log(1 - e^{-cx})$ and put $d_0(x; c) = f_0 - f_{\text{std}}$. Then d_0 is smooth with $d_0(0) = 0$. The standard product expansion is

$$\sum_{k \geq 1} f_{\text{std}}(k\varepsilon; c) = \frac{\pi^2}{6c\varepsilon} + \frac{1}{2} \log \frac{c\varepsilon}{2\pi} - \frac{c\varepsilon}{24} + O_R(\varepsilon^R) \quad (27)$$

for every fixed R . One proof uses Mellin inversion with kernel $\Gamma(v)\zeta(v)\zeta(v+1)$. Apart from $v = 1, 0, -1$, the poles at negative integers cancel by the trivial zeta zeros; moving the contour to any fixed negative line proves the stated finite-order version. Only this version is required here.

Euler–Maclaurin now gives

$$\log F_{\eta, K}(e^{-c\varepsilon}, T) = \frac{J(c)}{\varepsilon} + r_\eta \log \varepsilon + d_\eta(c) + \sum_{j \geq 0} \varepsilon^j H_j(c, z), \quad (28)$$

where

$$r_+ = \frac{1}{2}, \quad d_+(c) = \frac{1}{2} \log(c/(2\pi)), \quad (29)$$

$$r_- = 0, \quad d_-(c) = -\frac{1}{2} \log 2, \quad (30)$$

$$H_0(c, z) = \frac{M_1(c)}{2} + \frac{zM_2(c)}{2} - \frac{M_3(c)}{3}. \quad (31)$$

For $j \geq 1$ the explicit coefficient formula is

$$H_j = \int_0^\infty f_{j+1} dx - \frac{f_j(0)}{2} - \sum_{m=1}^{\lfloor j/2 \rfloor} \frac{B_{2m}}{(2m)!} f_{j-2m+1}^{(2m-1)}(0) + E_j(c), \quad (32)$$

where B_j are Bernoulli numbers, and superscripts on f_j in (32) mean x derivatives. For the Fermi case,

$$E_{2m} = 0, \quad E_j = -\frac{B_{j+1}}{(j+1)!} f_0^{(j)}(0) \quad (j \text{ odd}).$$

For the Bose case,

$$E_{2m} = 0, \quad E_1 = \frac{1/c - c}{24}, \quad E_j = -\frac{B_{j+1}}{(j+1)!} d_0^{(j)}(0) \quad (j \geq 3 \text{ odd}).$$

Thus only finitely many polynomials, integrals and endpoint derivatives are needed for any prescribed order. The formal coefficient functions can be extended to $k > K$ because their Gaussian tails are smaller than every algebraic order; this does not extend the original signed infinite product as an analytic object.

6 Fourier localization and the prefactors

For a central value of T , every $P_k(T)$ in the cutoff product is positive. On $q = q_0 e^{i\theta}$ the product ratio is a characteristic function of independent geometric or Bernoulli variables. Choose a fixed interval $as \leq k \leq bs$ with $0 < a < b$. Its parameters stay uniformly away from degeneracy, so for some positive constants d, d' ,

$$\frac{|F(q_0 e^{i\theta}, T)|}{F(q_0, T)} \leq \exp \left(-d \sum_{as \leq k \leq bs} (1 - \cos(k\theta)) \right) \quad (33)$$

$$\leq \exp(-d' \min(s^3 \theta^2, s)), \quad |\theta| \leq \pi. \quad (34)$$

For $|\theta| \leq c_0/s$, use the quadratic bound on $1 - \cos(k\theta)$. For $c_0/s \leq |\theta| \leq C_0/s$, use compactness and the nonzero limiting integral of $1 - \cos(xv)$. For $|\theta| \geq C_0/s$, the geometric-sum estimate for $\sum e^{ik\theta}$ gives a positive fraction of the interval length when C_0 is large enough. This covers the full circle, including possible rational angles.

Only $|\theta| \leq s^{-3/2} \log^2 n$ contributes at algebraic order. Write $\theta = us^{-3/2}$. In (28) this replaces c by $c' = c - iu\sqrt{\varepsilon}$. The linear phase from q^{-n} cancels because $J'(c) = -M_1(c) = -1$, and the quadratic phase is $-Vu^2/2$.

The density of $z = (T - n)/s$ has the finite expansion

$$\frac{e^{-z^2/2}}{\sqrt{2\pi}} \exp \left(\sum_{j \geq 1} \frac{(-1)^{j+1}}{j+2} \varepsilon^j z^{j+2} - \sum_{m \geq 1} \frac{B_{2m}}{2m(2m-1)} \varepsilon^{4m-2} \right). \quad (35)$$

It is obtained directly from the Gamma density and Stirling's formula; in particular its first correction is $\varepsilon z^3/3$, with no linear z term when centering at n .

At the saddle, the constant product amplitude is

$$e^{H_0} = \exp(1/2 - M_3/3) \exp(zM_2/2).$$

Integration against the normal density contributes $\exp(1/2 - M_3/3 + M_2^2/8) = E_0$. The Fourier integral contributes $(2\pi s^3 V)^{-1/2}$. The Euler-Maclaurin boundary contributes $\sqrt{c/(2\pi s)}$ for Bose and $1/\sqrt{2}$ for Fermi. Their product is exactly (6), with the powers n^{-1} and $n^{-3/4}$. Omitting the Gamma tilt or the cubic Stirling-transform contribution would change the amplitudes.

7 A finite coefficient algorithm at every order

Let U and Z be independent Gaussian variables with

$$U \sim N(0, V^{-1}), \quad Z \sim N(\alpha, 1), \quad \alpha = M_2/2.$$

Use $\sqrt{\varepsilon}$ as a formal variable and set $\Delta = -iU\sqrt{\varepsilon}$. Define the formal exponent

$$\begin{aligned} \mathcal{Q}(\varepsilon, U, Z) = & \sum_{r \geq 3} \frac{J^{(r)}(c)}{r!} (-iU)^r \varepsilon^{r/2-1} + d(c + \Delta) - d(c) \\ & + H_0(c + \Delta, Z) - H_0(c, Z) + \sum_{j \geq 1} \varepsilon^j H_j(c + \Delta, Z) \\ & + \sum_{j \geq 1} \frac{(-1)^{j+1}}{j+2} \varepsilon^j Z^{j+2} - \sum_{m \geq 1} \frac{B_{2m}}{2m(2m-1)} \varepsilon^{4m-2}. \end{aligned} \quad (36)$$

The complete coefficient prescription is

$$\boxed{a_j = [\varepsilon^j] \mathbb{E} \exp(\mathcal{Q}(\varepsilon, U, Z))}. \quad (37)$$

For fixed j , this requires only finite truncations of (11), (26), (32) and (36), followed by Gaussian moments. Odd half-integer powers integrate to zero because their integrands are odd in U . This proves that the expansion scale is $n^{-1/2}$ with no extra logarithmic factors.

7.1 Uniform remainder control

The cutoffs in largest part and Gamma variable have already been removed at relative precision in Section 4. Bound (34) is uniform in the remaining Gamma window. In the local product, the positive-root remainder (23) allows arbitrarily many terms to be retained. After removing the Bose logarithmic endpoint, Euler–Maclaurin remainders are controlled by integrals of derivatives; these are finite and at most polynomial in $|z|$, uniformly in a fixed complex neighborhood of c .

A further detail is important. The fixed-total profile majorant by itself does not bound the integrand $F(q_0, T)$, which includes all total sizes. Its joint Gaussian envelope follows instead from the local product expansion. Uniformly for $|z| \leq \log^2 n$,

$$\log F(q_0, T) = sJ + r_\eta \log \varepsilon + d_\eta + H_0(c, z) + O(\varepsilon(1 + z^2)). \quad (38)$$

For completeness, (22) gives on the cutoff range

$$\log R_k + x^2/2 = \varepsilon \ell_1 + O(\varepsilon^2(1 + z^2)P(x)), \quad P(0) = 0,$$

for a fixed polynomial P . The endpoint factor follows from the vanishing of the root power-sum polynomials at $k = 0$. The outer logarithm's linear derivative times P and quadratic derivative times ℓ_1^2 are integrable even in the Bose case. Summation over the mesh multiplies this bound by s ; Gaussian decay in x controls all polynomial factors. Endpoint corrections have the same bound, proving (38).

Because H_0 is affine in z , multiplication by the exact Gamma density gives, after extracting common scale factors, an upper bound

$$C_1 \exp\{-z^2/2 + \alpha z + O(\varepsilon(1 + |z|^3))\} \leq C_2 e^{-z^2/4}.$$

The error is absorbed since $\varepsilon \log^2 n \rightarrow 0$. Together with (34), this supplies a joint Gaussian envelope in u, z . Finite Taylor remainders can therefore be bounded by a fixed polynomial in $|u| + |z|$ times the target power of ε , multiplied by $e^{-d_2 u^2 - d_3 z^2}$ for positive constants d_2, d_3 . The resulting integrals are bounded independently of n . Expanding a few extra orders if necessary and integrating the next odd half-power, which vanishes exactly, gives an $O_L(\varepsilon^{L+1})$ remainder. No logarithmic cutoff losses remain. This justifies (37) at every fixed order and completes the proof of Theorem 1.1.

8 First corrections and executable calculations

Define the cumulant integrals

$$K_j = \int_0^\infty x^j \rho(1 + \eta \rho) dx.$$

Then the first product correction is the quadratic polynomial

$$\begin{aligned} H_1(z) &= e_\eta + \frac{3M_2}{4} - \frac{3M_4}{8} + \frac{K_2}{8} - \frac{K_4}{6} + \frac{K_6}{18} \\ &\quad + z \left(-\frac{M_1}{2} + \frac{2M_3}{3} + \frac{K_3}{4} - \frac{K_5}{6} \right) + z^2 \left(-\frac{M_2}{2} + \frac{K_4}{8} \right), \\ e_+ &= -\frac{c}{24} - \frac{5}{24c}, \quad e_- = \frac{c}{24}. \end{aligned} \tag{39}$$

Let $A_0(c, z) = e^{d(c) + H_0(c, z)}$, with c left variable while differentiating. The Fourier correction is

$$D(z) = -\frac{(A_0)_{cc}}{2VA_0} + \frac{J'''(A_0)_c}{2V^2A_0} + \frac{J''''}{8V^2} - \frac{5(J''')^2}{24V^3}. \tag{40}$$

Consequently

$$a_1 = \mathbb{E}_{Z \sim N(\alpha, 1)} [H_1(Z) + Z^3/3 + D(Z)]. \tag{41}$$

This provides an independent short route to the first coefficient. The script `first_correction.py` evaluates it by one-dimensional integrals and exact Gaussian moments.

The separate script `generic_second_correction.py` implements (37) through a_2 without using the hand-expanded (39) or (40). It uses the displayed root polynomials to construct the ℓ_j , cumulants, and all needed c derivatives. It reproduces both a_1 values and gives the a_2 values in the table. Higher orders use the same finite algorithm; the supplied executable is tested through second order rather than advertised as a fully optimized arbitrary-order software package.

9 Controlled smooth inverses

The sequence itself has only integer arguments. We first specify the sense of a smooth inverse. Let $G(x)$ be a smooth realization of its full logarithmic asymptotic expansion, with termwise differentiable asymptotics. Such a realization is obtained by multiplying successive terms by fixed smooth cutoffs at sufficiently rapidly increasing radii, chosen inductively to control finitely many additional derivatives at each step.

If exact interpolation is desired, choose a smooth bump ϕ supported in $(-1/3, 1/3)$ with $\phi(0) = 1$ and define

$$\log F(x) = G(x) + \sum_{\substack{m \text{ integer} \\ m \text{ sufficiently large}}} (\log B(m) - G(m)) \phi(x - m). \tag{42}$$

The supports are disjoint, and the correction coefficients and all their bump derivatives are smaller than every inverse power by Theorem 1.1. Thus $F(m) = B(m)$, F is positive, and $(\log F)'(x) = \log x + O(x^{-1/2})$. It is eventually strictly increasing. All such realizations have the same inverse expansion to every displayed order. No canonical analytic continuation is asserted.

Let

$$\log \left(1 + \sum_{j \geq 1} a_j x^{-j/2} \right) = \sum_{j \geq 1} b_j x^{-j/2}.$$

Stirling's formula gives

$$\log F(x) = x(\log x - 1) + C\sqrt{x} + \alpha_0 \log x + d_0 + \sum_{j \geq 1} \beta_j x^{-j/2}, \quad (43)$$

where $\alpha_0 = 1/2 - p$, $d_0 = \log(A\sqrt{2\pi})$, $\beta_1 = a_1$, and $\beta_2 = b_2 + 1/12$. For $Y \rightarrow \infty$, put

$$L = \log Y, \quad N = \frac{L}{W(L/e)}, \quad \ell = \log N,$$

with the positive real branch of Lambert W .

Theorem 9.1. *The smooth inverse admits a finite-order expansion of the form*

$$F^{-1}(Y) = N + \sum_{j \geq -1} N^{-j/2} r_j(1/\ell),$$

where each r_j is a polynomial in $1/\ell$, computed recursively from (43). In particular,

$$F^{-1}(Y) = N - \frac{C\sqrt{N}}{\ell} - \alpha_0 - \frac{d_0}{\ell} + \frac{C^2}{2\ell^2} - \frac{C^2}{2\ell^3} + O\left(\frac{1}{\sqrt{N}\ell}\right). \quad (44)$$

The constant shifts are $+1/2$ for unrestricted partitions and $+1/4$ for distinct parts.

Proof. Substitute $x = N + \Delta$ in (43) and use $N(\log N - 1) = L$. At each new power of $N^{-1/2}$, the unknown coefficient occurs multiplied by ℓ , so it is determined uniquely by division by ℓ . Taylor expansion first cancels the $\sqrt{N}$ term, then the constant term, yielding (44). All remaining operations preserve polynomials in $1/\ell$.

For quantitative control, substitute a finite inverse truncation X into the logarithmic expansion and compute its residual to one more order. If $\log F(X) - L = O(N^{-r/2})$, the mean-value theorem and $(\log F)' \sim \log N$ give inverse error $O(N^{-r/2}/\log N)$. This proves each prescribed finite order; it does not invert only the leading equivalent. $\square$

A useful next term can be obtained without a large expanded polynomial. Define

$$\begin{aligned} a &= -C/\ell, & b &= -\alpha_0 - d_0/\ell + C^2/(2\ell^2) - C^2/(2\ell^3), \\ d &= -\frac{(a + C/2)b - a^3/6 - Ca^2/8 + \alpha_0 a + \beta_1}{\ell}. \end{aligned} \quad (45)$$

Then $N + \sqrt{N}a + b + d/\sqrt{N}$ has error $O(1/(N\ell))$.

For the integer threshold $\tau(Y) = \min\{n : B(n) \geq Y\}$, eventual monotonicity follows from the main expansion. With an exact increasing interpolant as in (42), $\tau(Y) = \lceil F^{-1}(Y) \rceil$ for sufficiently

large Y . A finite approximation $X(Y)$ with a valid error bound $e(Y)$ gives only the rigorous envelope

$$\lceil X(Y) - e(Y) \rceil \leq \tau(Y) \leq \lceil X(Y) + e(Y) \rceil. \quad (46)$$

Exact rounding follows when the endpoints coincide, not unconditionally from a finite asymptotic formula.

[write] **Added 2 October 2026, batch 77: relation to the repository's inversion apparatus.** No novelty is claimed for the inversion method; it is an instance of ProveIt's transseries volumes. The core $N = L/W(L/e)$ solves $x(\log x - 1) = L$: it is the unbalanced core `t2:eq:unbalanced-core` of *Combinatorial Transseries and Their Inverses* [6] with $\kappa = 1$, $\lambda = -1$, equivalently Proposition `p0:prop:factorial-core` of [5] with $\kappa = 1$, $d = -1$ and the principal branch W_0 . The recursion in the proof of Theorem 9.1, which removes the terms $C\sqrt{x} + \alpha_0 \log x + d_0 + \sum \beta_j x^{-j/2}$ order by order, is a local reversion around that core in the sense of `t2:prop:scaled` of [6] (additive displacement); its first correction, of order $\sqrt{N}/\log N$, has the same shape as the involution inverse of [6], chapter `ct:ch:involutions`. The mean-value step is the residual-to-root conversion `p0:thm:backward-error`. The interpolant (42) is an admissible interpolation in the sense of `p0:def:three-inverses`, so $\tau(Y) = \lceil F^{-1}(Y) \rceil$ is `p0:thm:staircase`, part (1), and the envelope (46), with its separation proviso, is part (2); the independence of the inverse expansion from the choice of realization is the point of `p0:rem:three-differ` [5]. What is specific to these sequences is the input: C_η , α_0 , d_0 and the β_j , and the constant shifts $+1/2$ and $+1/4$. The same applies to the chromatic inverse at the end of Section 11.

10 Exact checks and reproducibility

The program `exact_check.py` computes both sequences by positive Touchard polynomial arithmetic followed by one exact alternating factorial evaluation. Let $Q_k(t) = \sum_j S(k, j)t^j$. Compute

$$H_{+,n}(t) = [q^n] \prod_{k=1}^n (1 - q^k Q_k(t))^{-1}, \quad H_{-,n}(t) = [q^n] \prod_{k=1}^n (1 + q^k Q_k(t)).$$

If $h_{n,j}$ are their coefficients, then $B_\eta(n) = \sum_j (-1)^{n-j} h_{n,j} j!$. All intermediate $h_{n,j}$ are nonnegative. For a fixed profile the sum of its coefficients is at most the Bell number $\text{Bell}(n)$: its blockwise set partitions inject into set partitions of a fixed n -set. Hence

$$h_{n,j} \leq p(n) \text{Bell}(n) \leq 2^n n^n.$$

The binary base in the Kronecker encoding is chosen above this bound, so no polynomial digit can carry. All sequence values are exact Python integers; cancellation occurs only in the final exact evaluation.

Separate runs at maximum sizes 150 and 300 agree on every overlapping value, and all 21 initial OEIS terms used in the check agree. Let

$$R_\eta(n) = \frac{B_\eta(n)}{A_\eta n! n^{-p_\eta} e^{C_\eta \sqrt{n}}}, \quad T_\eta(n) = n^{3/2} \left(R_\eta(n) - 1 - \frac{a_{\eta,1}}{\sqrt{n}} - \frac{a_{\eta,2}}{n} \right).$$

Representative exact-data residuals are

n	$R_+(n)$	$T_+(n)$	$R_-(n)$	$T_-(n)$
100	0.9694277210	-0.00301198	1.0230946532	0.04210253
150	0.9750093079	-0.00538650	1.0188567728	0.05158749
200	0.9783435022	-0.00559355	1.0163242684	0.04202347
250	0.9806213569	-0.00557333	1.0146001519	0.04176652
300	0.9823040643	-0.00553406	1.0133277008	0.04126194

The slower and mildly oscillatory finite-size behavior in the distinct case does not enter the proof. These checks support, but do not replace, the analytic argument.

The verification script also tests the inverse at $Y = B_\eta(300)$. Formula (44) differs from 300 by approximately 0.00350453 and 0.00380904, respectively. Including (45) changes these to approximately -0.0000155943 and 0.0000217153 . These are numerical residual checks, not certified threshold bounds.

The reproducibility package contains the TeX source, PDF, proof source, exact arrays, numerical outputs, verification scripts and review record. Its `README.md` specifies the short replay and full exact replay. The second-order executable uses Python with `sympy` and `mpmath`; no proprietary algebra system is required.

[write] **Added 2 October 2026, batch 77: the shipped files.** In ProveIt the reproducibility package is this report's directory, with a different layout: the programs and `build.sh`, `replay.sh` are in `code/`, the exact arrays, numerical outputs, logs and requirements in `data/`, and `proof.md`, `tutte_axis_extension.md` and the three audit notes (as `independent_audit-*.md`) beside this article; the audit program and its outputs carry the prefix `independent_audit-`. The delivered PDF and the two hash ledgers (`manifest.json` and `independent_audit/reviewed_hashes.json`) are not shipped; the ledgers were verified at intake, and all three survive in the archive of commit `096ee7b87`. The scripts expect their JSON inputs beside them, as in the delivery; the report `README` gives rerun commands that work with the shipped names. At intake (2 October 2026) the short replay was run on a copy: the three recomputed JSON files equalled the recorded ones up to line endings, and the two first-correction routes agreed within 10^{-45} . At the write the independent audit program was also run on a copy and passed. The exact $n \leq 300$ recomputation and the PDF rebuild were not rerun.

11 A fixed positive chromatic parameter

The proof also treats the natural chromatic specialization $H_v(G) = (-1)^{|V(G)|} \chi_G(-v)$ for fixed $v > 0$. Let $B_{\eta,v}(n)$ sum $H_v(K_\lambda)$ over the same two partition families. This section asserts uniformity only when v ranges over a fixed compact subset of $(0, \infty)$.

Corollary 11.1. *The constants A_η, C_η, p_η in Theorem 1.1 are unchanged, and*

$$B_{\eta,v}(n) = \frac{\Gamma(n+v)}{\Gamma(v)} A_\eta n^{-p_\eta} e^{C_\eta \sqrt{n}} \left(\sum_{j=0}^L a_{\eta,j}(v) n^{-j/2} + O_L(n^{-(L+1)/2}) \right). \quad (47)$$

The coefficients have the same finite construction, and

$$a_{\eta,1}(v) = a_{\eta,1}(1) + (v-1)M_2/2.$$

Proof. Replace (9) by the exact identity

$$H_v(K_\lambda) = \frac{1}{\Gamma(v)} \int_0^\infty e^{-t} t^{v-1} \prod_i P_{\lambda_i}(t) dt.$$

The normalized expectation uses $T \sim \text{Gamma}(n + v, 1)$. In the formal exponent (36), replace the Gamma part by

$$\sum_{j \geq 1} \frac{(-1)^{j+1}}{j+2} \varepsilon^j Z^{j+2} + (v-1) \log(1 + \varepsilon Z) - \sum_{r \geq 1} \frac{(-1)^{r+1} \mathbf{B}_{r+1}(v)}{r(r+1)} \varepsilon^{2r},$$

where $\mathbf{B}_r(v)$ is a Bernoulli polynomial. This follows from generalized Stirling expansion of $\Gamma(n+v)$. The only first-order change is $(v-1)Z$, with tilted mean $(v-1)M_2/2$.

On the central interval $t/n \in [1/2, 2]$, the extra factor t^{v-1} is uniformly comparable to n^{v-1} , and $\Gamma(n+v) \asymp n!n^{v-1}$. All majorants remain uniform. For the lower signed tail, including $0 < v < 1$, do not bound t^{v-1} pointwise near zero. Instead $e^{-t}(t+M)^n$ is increasing on $[0, n/2]$ for large n , and its endpoint bound is multiplied by $\int_0^{n/2} t^{v-1} dt = (n/2)^v/v$. This gives the same exponential suppression uniformly for v bounded away from zero. The upper tail is unchanged by fixed powers.

Finally H_v is positive: chromatic coefficients have alternating nonnegative signs. Deletion-contraction gives $H_v(G+e) = H_v(G) + H_v(G/e)$, hence edge monotonicity. With $m = n - L$,

$$H_v(K_{L,1,\dots,1}) = \frac{\Gamma(m+v)}{\Gamma(v)} (m+v)^L.$$

Dividing by $\Gamma(n+v)/\Gamma(v)$ gives, by the same logarithmic inequality as (20), the bound $\exp(-L(L-1)/(2(n+v-1)))$. The far-part tail and every local estimate therefore transfer to compact positive v sets. $\square$

For the inverse, normalize (47) relative to $n!n^{v-1}$. In (43) this changes the constants to

$$\alpha_0 = v - \frac{1}{2} - p_\eta, \quad d_0 = \log \left(\frac{A_\eta \sqrt{2\pi}}{\Gamma(v)} \right).$$

The same Lambert- W core and recursion apply. The Gamma-ratio series affects β_2 and higher terms, but not $\beta_1 = a_{\eta,1}(v)$.

12 Further questions

The complete algebraic expansion leaves several substantial questions. First, one can study the late growth of its coefficients and the exponentially small sectors arising from noncentral coefficient contours; the present theorem does not identify these sectors or a summation prescription. Second, the dominant weighted partition model suggests limit laws for the largest part and for profile statistics under orientation weighting. Turning this into uniform distributional results requires additional marked-parameter estimates. Third, a chromatic parameter growing with n could change the Gamma tilt and the saddle regime; compact-parameter arguments do not cover that transition. Finally, explicit effective constants in the full remainder would turn the inverse threshold envelope into a practical certified numerical algorithm.

[write] **Added 2 October 2026, batch 77.** As of this write no report of the collection and no transseries volume of Provelt answers any of these questions.

References

- [1] The OEIS Foundation, [A372395](#), total acyclic orientations over complete multipartite graphs indexed by integer partitions. Accessed 1 October 2026.
- [2] The OEIS Foundation, [A370613](#), the corresponding distinct-part partition sum. Accessed 1 October 2026.
- [3] R. P. Stanley, Acyclic orientations of graphs, *Discrete Mathematics* **5** (1973), 171–178. [doi:10.1016/0012-365X\(73\)90108-8](#).
- [4] Z. Sun, Saddle-Point Asymptotics for Chromatic and Tutte Polynomial Evaluations of Complete Multipartite Graphs, [arXiv:2605.04006v1](#) (2026), especially Sections 2, 7–9 and 11.
- [5] [write] ProveIt, *Transseries: the polynomial-logarithmic calculus, series reversal at infinity, and inversion*, `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`; `p0:prop:factorial-core`, `p0:thm:backward-error`, `p0:def:three-inverses`, `p0:rem:three-differ`, `p0:thm:staircase`.
- [6] [write] ProveIt, *Combinatorial Transseries and Their Inverses*, `Analysis/Transseries/docs/series-and-transseries/Combinatorial_Transseries_Inverses/Combinatorial_Transseries_Inverses.tex`; `t2:eq:unbalanced-core`, `t2:prop:scaled`, chapter `ct:ch:involutions`.